

There is huge economic value in solving the problem of sentiment analysis in text. Companies that sell products hugely depend on the customer reviews. If there are tools and mechanisms in place by which they are able to analyse the customer's sentiments, the sellers can get a granular look at the issues that their product is facing. This removes a lot of the human labour involved in tasks such as this, and also optimises the process for a myriad of other features such as product recommendations, increasing user acquisition while decreasing user attrition rate. For social media companies, natural language understanding is crucial in identifying posts with abuse, hate-speech, inciteful content and spam.

Types of sentiment analysis for text based data

Sentiment analysis of text is a broad based term that covers many different techniques used for specific types of sentiment analysis. In general, it focuses on understanding the polarity of a given piece of text, i.e., positivity, negativity or neutrality conveyed in the text. However, there can be more depth to understanding the sentiments conveyed in the text.

Graded sentiment analysis: Graded sentiment analysis takes the generalised search for polarity in text to a deeper level. For example, let us say A and B are two sample texts. It is possible that both convey a negative sentiment. But it is also possible that A is more negative than B. In such a case, graded sentiment analysis can be used to not only understand the polarity of a piece of text but also grade it for the level of polarity it holds, such as:

- Very negative
- Negative
- Somewhat negative
- Somewhat neutral
- Neutral
- Very neutral
- Somewhat positive
- Positive
- Very positive

Emotion and sentiment analysis: In some cases, understanding the polarity of a text is not enough. You may also like to understand the specific emotions that are being conveyed in it. For instance, a text can be negative but, specifically, it may express more sadness than anger or vice versa. This can be crucial when trying to detect hate speech on social media or to understand how your product made your customer feel on an e-commerce platform.

Emotion detection in sentiment analysis can be done to understand emotions expressed in text such as:

- Anger (Negative)
- Sadness (Negative)
- Joy (Positive)
- Happiness (Positive)
- Gratitude (Neutral)
- Honesty (Neutral)

Aspect based sentiment analysis: In certain cases, it is not enough to understand the polarity and emotion in a piece of text; we also require some context in the form of aspects or features. For instance, it isn't enough for a seller selling a washing machine on Amazon to know whether a user review contained negativity and anger. The seller would instead like to know the specific features or aspects of the product that actually caused the anger or frustration in the customer. Was it the new 'timer system' or the 'product size' or the 'product design' that caused the issue for the customer? These are all aspects of the washing machine, and if the computer is able to understand not just the sentiments of the customer but also the reasons for the sentiment, this can provide the seller with a lot more context. Amazon reviews are one of many use cases for aspect based sentiment analysis.

Multilingual sentiment analysis: Things can get very complicated very fast for machine learning algorithms when you introduce more than one language into the mix. Every language has its own set of rules and methods of expression. There may be overlap in certain regards but every language brings its own unique baggage. For instance, both Spanish and French have their roots in Latin. However, sentence construction and the spelling of words have their own methods and rules. How is a computer supposed to understand the sentiments of a text when a bilingual user uses some words in French and other words in Spanish? In India we tend to write 'Hindi' and 'Urdu' using the English alphabet. From texting to making comments on YouTube, this is a very common practice in India. To understand such a piece of text can be extremely challenging. This is where multilingual sentiment analysis comes in. Breakthroughs are being made every day but there is a lot that still needs to be done in this particular field.

How to use NLP and ML for sentiment analysis of text based data

Now that we have a clear understanding of what NLP is and why it is a crucial field of research in AI and ML, we shall now shift our focus to discussing how you as a beginner in NLP and ML can get started by building your own language understanding models.

In sentiment analysis using AI, building a language understanding model requires three things:

1. A specific problem to solve
2. A relevant data set